Division

Proposition 1. *Given any two integers $a, b \in \mathbb{Z}$ with $b \neq 0$, there are unique integers q and r (called quotient and remainder respectively) s.t. both the properties below hold:*

(1) $0 \leq r < |b|$
(2) $a = qb + r$

Proof. (Uniqueness):

- Suppose that the pair (q, r) is not unique. Therefore there is a distinct pair (q', r') satisfying the same properties above.
- $a = qb + r = q'b + r'$. Hence subtracting we get:
  - $(q - q')b = r' - r$, hence $|q - q' \mid |b| = |r' - r| \leq |r'| + |r| < 2|b|$.
  - Hence $|q - q'| < 2$. Note that $|q - q'| \geq 0$, hence $|q - q'|$ is 0 or 1.
    - If it is 0, $q = q'$, left hand side is 0 and hence $|r - r'| = 0$, hence $r = r'$. Hence a contradiction to their being distinct.
    - If it is 1, $|b| = |r' - r|$. Either $r' > r$, or $r' < r$ or $r' = r$.
      - If $r' = r$: $|b| = 0$, not possible.
      - If $r' > r$: $|b| = |r' - r| = r' - r < r'$ since $0 \leq r$. But $r' < |b|$ by assumption, not possible.
      - If $r' < r$: ditto, changing the roles of r' and r.

(Existence): □

- Consider the set $S = \{a - nb \mid n \in \mathbb{Z}, a - nb \geq 0\}$.
- $S \neq \phi$: If $a \geq 0$, then $n = 0 \in S$ since $a - 0 \cdot b = a \geq 0$. If $a < 0$, $n = -sign(b)|a| \in S$ since

$$a - (-sign(b) \cdot |a|b) = a + |a||b| \geq a + |a| \geq 0.$$

- Hence it has a minimum element, say $m = a - n_0 b$ for some $n_0 \in \mathbb{Z}$. We claim $m < |b|$.
  - If not: we have $m - |b| \geq 0$ but $m - |b| = a - (n_0 + sign(b))b$ and hence is a strictly smaller element in S.
- Since $m \in S$, $m \geq 0$ and hence $a = n_0 b + m$ with $0 \leq m < |b|$. Hence $q = n_0$, $r = m$ satisfy the given property as required.

Definition 2. For $a, b \in \mathbb{Z}$, we say b divides a, written $b \mid a$, when the unique remainder above is 0, or in other words there is a $q \in \mathbb{Z}$ s.t. $a = qb$. For $u, v, n \in \mathbb{Z}$ the notation "$u \equiv v \pmod{n}$" (read u is congruent to v modulo n) will stand for the statement $n \mid (u - v)$, i.e. n divides $u - v$.

Exercise 3. Show that:

(1) Show that if $m \mid n$ and $n \neq 0$, then $|m| \leq |n|$.
(2) $u \equiv v \pmod{n}$ and $u' \equiv v' \pmod{n}$ then $u + u' \equiv v + v' \pmod{n}$, $-u \equiv -v \pmod{n}$ and $uv \equiv u'v' \pmod{n}$. In this respect, equivalence $\pmod{n}$ behaves like identity.
(3) Show that $uv \equiv u'v \pmod{n}$ does not always imply that $u \equiv u' \pmod{n}$ by giving an example. Thus the cancellation law for multiplication fails in this case!

Exercise 4. Compute

(1) the remainder when 7 divides 2^{2025}.
(2) the remainder when 99 divides 10^{2025}.

Definition 5. Given two integers m, n with at least one of them non-zero, the set $S = \{d \text{ such that } d \mid m, d \mid n, d > 0\}$ is finite (being bounded by the non-zero element) and hence has a maximum. We call $\max S$ as the greatest common divisor of m, n denoted $\gcd(m, n)$.

Definition 6 (Euclid's Division Algorithm)**.** For $u, v > 0$ the following gives an algorithm to compute $\gcd(u, v)$ where we assume, without loss of generality, that $v \leq u$.

(1) If $v \mid u$ output v, otherwise ...
(2) ... write $u = qv + r$ with $0 < r < n$. Output $\gcd(v, r)$ which in turn is computed by going to step 1.

Exercise 7. Show that:

(1) $\gcd(m, n) = \gcd(m, n + sm)$ for any integer s.
(2) $\gcd(m, n) = \gcd(m, -n) = \gcd(n, m)$.
(3) The $\gcd(v, r)$ in the step 2 of the algorithm computes the $\gcd(u, v)$ as required.
(4) The algorithm terminates in finitely many steps.

Proposition 8. *For any integers u, v not both zero, we have integers m, n such that $mu + nv = \gcd(u, v)$.*

Proof. We induct on the number of times the step 1 is repeated in the algorithm above when computing $\gcd(u, v)$.

(Base case): If the step 1 occurs only once, $v \mid u$ and hence $\gcd(u, v) = v$. In this case $0 \cdot u + 1 \cdot v = v$ and hence $(m, n) = (0, 1)$ do the job.

(Induction step):

Assume that if for some u, v, the number of time step 1 occurs in computation of $\gcd(u, v)$ is k, then $mu+nv = \gcd(u, v)$. [Induction hypothesis]. We need to show that the same also holds when the number of times step 1 occurs in the computation of gcd is $k + 1$.

The first time step 2 occurs, we have the computation: $\gcd(u, v) = \gcd(v, r)$ where $u = qv + r$. Now for computation of $\gcd(v, r)$ step 1 will occur one fewer time, and hence, by induction hypothesis, $m'v + n'r = \gcd(v, r) = \gcd(u, v)$. Replacing $r = u - qv$, we get $m'v + n'u - n'qv = \gcd(u, v)$, that is

$$n'v + (m' - n'q)v = \gcd(u, v)$$

and we can take $m = n'$ and $n = m' - n'q$. This completes the induction step and we are done. □

Exercise 9. Show that the following two sets are equal:

$$\{mu + nv \mid m, n \in \mathbb{Z}\} = \{r \cdot \gcd(u, v) \mid r \in \mathbb{Z}\}$$

Exercise 10. Compute the pair of integers satisfying the relation below in each case, or if none exist state so with reasons.

(1) $7m + 97n = 12$
(2) $12m + 21n = 2$

Exercise 11. Describe all the pairs m, n such that $77m + 121n = 22$.

Definition 12. A number $p \in \mathbb{N}$is said to be prime if $d \mid p$ for some $d \in \mathbb{N}$ then $d = 1$ or $d = p$.

Corollary 13. *Assume p is a prime number and $a, b \in \mathbb{Z}$. If $p \mid ab$ then $p \mid a$ or $p|b$ (or both).*

Proof. Take such p, a, b with $p \mid ab$ and p prime. If $p \mid a$, we are done. If $p \nmid a$ then $\gcd(p, a) = 1$ by definition. Hence by previous proposition, there exists $m, n \in \mathbb{Z}$ such that

$$mp + na = 1 \Rightarrow mpb + nab = b.$$

Now note that $mpb \equiv m \cdot 0 \cdot b \equiv 0 \pmod p$ and $nab \equiv n \cdot (ab) \equiv n \cdot 0 \equiv 0 \pmod p$ (since $ab \equiv 0 \pmod p$ by assumption). Hence $b = mpb + nab \equiv 0 + 0 \equiv 0 \pmod p$. □

Exercise 14. Show that for any set $X \subset \mathbb{Z}$ which contains at least one non-zero element,

$$S_X := \{d \text{ such that } d > 0, d \text{ divides every element of the set } X\}$$

is a finite set and hence $\gcd(X) := \max S_X$ makes sense and coincides with the definition above when X has two elements.

Exercise 15. Show that:

(1) for any $k \geq 1$, if $p \mid a_1 a_2 \cdots a_k$ where p is a prime, then $p \mid a_i$ for some i.
(2) for any $k \geq 1$, and any $a_1, a_2, \ldots, a_k \in \mathbb{Z}$ the following two sets are equal:

$$\{m_1 a_1 + m_2 a_2 + \cdots + m_k a_k \text{ such that } m_1, m_2, \ldots, m_k \in \mathbb{Z}\} = \{m \gcd(a_1, a_2, \ldots, a_k) \mid m \in \mathbb{Z}\}$$

where $\gcd(a_1, a_2, \ldots, a_k)$ was defined in the previous exercise.

Corollary 16 (Fundamental Theorem of Arithmetic)**.** *For any $n \in \mathbb{N}$ there is a factorization $n = p_1 p_2 \cdots p_r$ for some $r \geq 0$ (for $r = 0$, RHS is 1 by convention. Further, it is unique in the following sense:*

$$p_1 p_2 \cdots p_r = p'_1 p'_2 \cdots p'_{r'} \textit{ then } r = r' \textit{ and LHS is a permutation of the RHS.}$$

Proof. (Existence): By induction on n. (Base case) $n = 1$, then $r = 0$ (Induction step): Suppose the result holds for $n \leq N$. Now assume $n = N + 1$. Then if n is prime, $r = 1$ and $p_1 = n$. If n is not prime, it has at least one factor $1 < d < n$ and hence $n = dq$ but then $1 < q < n$ (easy exercise!), that is $d, q \leq N$. Hence by induction hypothesis, both d, q are product of primes. But then so is $dq = n$!

(Uniqueness): Inducting on $\min(r, r')$. (Base case) If $\min(r, r') = 0$, then $n = 1$ (empty product), and therefore if $p_1 p_2 \cdots p_r = 1$ then $p_i \leq 1$ which is not possible. (Induction step): Suppose result holds for $\min(r, r') = N$. Now assume $\min(r, r') = N + 1 \geq 1$. Then $p_1$divides LHS (by definition) hence, RHS, and hence at least one of p_i. Now only non unit factors of p_i are 1 and p and since 1 is not possible, $p_i = p$. Now cancelling $p_i = p$ on both sides, we get $r - 1$ primes on LHS and $r' - 1$ on RHS. By induction, $r - 1 = r' - 1$ and hence $r = r'$. Also by induciton the multi-set $\{p_2, \ldots, p_r\}$ is a permutation of $\{p'_2, \ldots, p'_{r'}\}$ as required and we are done. □

Exercise 17. Show that if $\gcd(a, b) = d$ and $m \in \mathbb{Z}$ is such that $m \mid a$ and $m \mid b$ then $m \mid d$.

Exercise 18. Show that if $ma + nb = p$ for some prime p and some integers m, n, a, b. Then there are integers m_k, n_k such that $m_k a^k + n_k b^k = p^k$.